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# Distributed Temporal Neural Architectures for Scalable Cognitive Modeling

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# Distributed Temporal Neural Architectures for Scalable Cognitive Modeling

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**ABSTRACT** Modeling human cognitive dynamics from longitudinal time-series data is a multi-scale temporal problem: cognitive states evolve over fast attentional shifts (seconds), task-level mood variation (minutes), and slow fatigue accumulation (hours to days). Scaling cognitive modeling to large participant cohorts and continuous sensing infrastructures further demands distributed computation. Existing temporal neural architectures are designed for single-machine execution and apply a uniform temporal resolution, failing to capture multi-scale cognitive dynamics; standard data-parallel distribution additionally violates the causal temporal ordering intrinsic to cognitive sequences. We present D-MSTCN (Distributed Multi-Scale Temporal Cognitive Network), a neural architecture that simultaneously addresses multi-scale temporal modeling and distributed scalable execution. D-MSTCN comprises three parallel branches of dilated temporal convolutions with scale-specific inductive biases — short, medium, and long — fused via a learned Cross-Scale Attention Gate (CSAG) that dynamically weights temporal horizons. Distributed execution is realised through a Scale-Aware Partitioner (SAP) coordinated by a Temporal Consistency Protocol (TCP) that enforces causal ordering with a bounded-staleness guarantee. A lightweight per-subject Personalization Adapter conditions the global model on individual cognitive baselines without per-subject retraining. We evaluate D-MSTCN on two longitudinal datasets — StudentLife and DAIC-WOZ — alongside seven baselines. D-MSTCN achieves the highest macro F1 on both datasets while delivering near-linear throughput scaling (efficiency  $\eta \geq 0.81$ ) from 1 to 8 compute nodes and reducing per-epoch training time by 57% at 8 nodes. To our knowledge, D-MSTCN is the first neural architecture to jointly address multi-scale temporal inductive bias and distributed temporal coherence for cognitive state modeling.

**INDEX TERMS** Cognitive state classification, cross-scale attention, dilated temporal convolutional networks, distributed deep learning, multi-scale temporal modeling, personalization.

## I. INTRODUCTION

### A. BACKGROUND AND MOTIVATION

**H**UMAN cognition is an inherently temporal phenomenon. Attention drifts and refocuses on timescales of seconds; task-related mood and working memory load evolve over minutes; fatigue and affective baselines shift over hours and days. Capturing this multi-scale temporal structure is essential for building accurate computational models of cognitive states, with significant applications in mental health monitoring, human–computer interaction (HCI), educational technology, and clinical screening.

Recent advances in passive, continuous sensing through smartphones, wristbands, and environmental sensors have made longitudinal cognitive data collection practical at scale.

Datasets such as StudentLife [1], which tracked 48 university students over a full academic term via smartphone sensors, and DAIC-WOZ [2], which recorded 189 clinical interview sessions for depression screening, exemplify a growing class of longitudinal cognitive datasets. As deployment scales from tens to hundreds and eventually thousands of participants across multi-site studies, the computational demands of temporal cognitive modeling grow commensurately. Wearable and edge sensing infrastructures further introduce the practical need to coordinate training across distributed compute nodes rather than centralising all data.

Yet the dominant temporal neural architectures — Long Short-Term Memory networks [3], Temporal Convolutional Networks (TCN) [4], and time-series Transformers [5], [6]

— were designed for single-machine execution. They apply a uniform temporal resolution across the entire sequence, failing to exploit the multi-scale inductive bias that cognitive dynamics demand. More critically, when distributed via standard data-parallel training [7], these models suffer gradient inconsistencies arising from violations of temporal causal ordering: partitioned batches drawn from different temporal positions in a longitudinal sequence yield gradients that are causally inconsistent when aggregated without temporal awareness.

## B. PROBLEM STATEMENT

We identify three distinct failure modes of existing approaches for scalable cognitive modeling. (P1) Single-scale temporal representation: standard temporal models — LSTMs, TCNs, and vanilla Transformers — apply a fixed temporal resolution to the entire input sequence. A single dilation rate or attention span cannot simultaneously capture fast attentional dynamics (effective on 2–10 second windows) and slow fatigue accumulation (effective on multi-hour windows). Models trained with a single temporal resolution systematically under-fit one of these timescales, producing inferior predictions in tasks where both fast and slow dynamics are informative [8].

(P2) Distributed training violates temporal causality: data-parallel distributed training partitions the dataset across nodes and applies all-reduce gradient synchronisation at each step. For temporal cognitive sequences, this is problematic: a gradient computed on timesteps  $t = 500$ –1000 on node 1 and another computed on  $t = 100$ –600 on node 2 are not exchangeable without introducing temporal inconsistency. Standard all-reduce averages these causally incompatible gradients, degrading convergence and model quality for sequential tasks.

(P3) Inter-subject variability at scale: cognitive signals exhibit substantial inter-subject variability — a physiological stress signature for one participant may resemble a rest signature for another. At population scale, a single global model cannot adequately fit all subjects, yet per-subject model retraining is prohibitively expensive and introduces privacy concerns.

Existing distributed learning frameworks including Federated Averaging (FedAvg) [9], PipeDream [10], and GPipe [11] address scalability but do not model multi-scale temporal structure and do not provide causal temporal consistency guarantees for longitudinal cognitive sequences.

## C. CONTRIBUTIONS

This paper makes five primary contributions:

- 1) **D-MSTCN architecture**: a three-branch parallel dilated temporal convolutional network with scale-specific dilation schedules — short ( $r \in \{1, 2, 4, 8\}$ ); medium ( $r \in \{8, 16, 32, 64\}$ ); long ( $r \in \{32, 64, 128, 256\}$ ) — and a Cross-Scale Attention Gate (CSAG) that dynamically fuses branch representations based on input context. This implements a neuroscientifically-grounded multi-scale inductive

bias for cognitive modeling, differing fundamentally from TimesNet’s FFT-based period discovery [8] and from pipeline-parallel frameworks (GPipe, PipeDream) which partition model layers rather than parallel temporal branches.

- 2) **Temporal Consistency Protocol (TCP)**: a gradient synchronisation mechanism for distributed D-MSTCN training that tracks parameter staleness per branch node, applies causal masking to gradients, and provides a bounded-staleness guarantee (Theorem 1) ensuring convergence within a neighbourhood of the synchronous optimum.
- 3) **Scale-Aware Partitioner (SAP)**: a communication–computation cost model and load-balancing algorithm that assigns D-MSTCN branches to compute nodes by minimising total synchronisation overhead.
- 4) **Personalization Adapter**: a lightweight per-subject FiLM conditioning [12] layer that adapts the global D-MSTCN model to individual cognitive baselines without per-subject retraining.
- 5) **Comprehensive evaluation**: we benchmark D-MSTCN against seven competitive baselines on two established cognitive datasets (StudentLife, DAIC-WOZ) and an extended EEG-based transfer experiment (SEED [13]).

Experimental results demonstrate that D-MSTCN achieves the highest macro F1 among all evaluated methods on both primary datasets while attaining scalability efficiency  $\eta \geq 0.81$  at 8 compute nodes and reducing per-epoch training time by 57% at 8 nodes versus single-node execution on the same architecture.

## D. PAPER ORGANIZATION

The remainder of this paper is organised as follows. Section II reviews related work. Section III presents the D-MSTCN architecture, TCP, and SAP, with full mathematical formulation. Section IV describes datasets, baselines, metrics, and the training protocol. Section V presents the main experimental results and scalability analysis. Section VI reports ablation studies. Section VII discusses findings and failure cases. Section VIII acknowledges limitations. Section IX concludes with future directions.

## II. RELATED WORK

### A. TEMPORAL NEURAL ARCHITECTURES

*Recurrent networks.* LSTM [3] and GRU [14] enable modeling long-range temporal dependencies through gated memory cells, but their sequential processing limits parallelism and distributed execution. For longitudinal cognitive sequences of 200–4000 timesteps, LSTM training becomes a sequential bottleneck.

*Temporal Convolutional Networks.* Bai et al. [4] demonstrated that dilated causal convolutional networks match or exceed LSTMs while enabling parallel processing across timesteps. The exponential growth of the receptive field with dilations  $r \in \{1, 2, 4, \dots, 2^k\}$  provides long-range dependency modeling at linear cost. However, standard TCN

applies a single uniform dilation schedule, a design choice that cannot simultaneously optimise for fast and slow cognitive dynamics.

*Transformer-based time-series models.* Transformers [15] were adapted for time-series by Autoformer [5], Informer [6], and PatchTST [16]. These leverage attention to capture long-range dependencies but at  $\mathcal{O}(T^2)$  or  $\mathcal{O}(T \log T)$  complexity, prohibitive for thousands of timesteps without windowing.

*Multi-period and multi-resolution models.* TimesNet [8] reshapes 1D time-series into 2D representations via FFT-based period detection and applies 2D convolution blocks. iTransformer [17] inverts Transformer attention to operate across the variate dimension. LSTNet [18], MICN [19], and SCINet [20] propose alternative multi-scale or hierarchical formulations. These approaches share critical limitations: (i) they are designed for centralised single-machine execution; (ii) their multi-scale mechanisms are learned data-adaptively rather than grounded in the physiological timescales of human cognition; (iii) none incorporates mechanisms to preserve temporal causal ordering under distributed execution.

## B. COGNITIVE STATE MODELING

The StudentLife dataset [1] has been extensively used for mental health prediction from passive sensing. Prior works have employed SVMs with handcrafted features [1], LSTM models over sensor windows [21], and attention-based architectures [22]. These methods operate on short windows (minutes), failing to capture the longitudinal evolution of stress over hours and days that our long-scale branch is designed to model.

The DAIC-WOZ benchmark [2] has driven substantial work in depression detection from acoustic features. Williamson et al. [23] applied random forests over eGeMAPS features. Deep learning approaches include LSTMs for acoustic sequences [24], temporal attention models [25], and multimodal Transformers [26]. These methods achieve competitive PHQ-8 classification accuracy but are uniformly single-machine and single-scale.

The SEED dataset [13] provides EEG-based emotion recognition and has been addressed with spectral CNNs [27] and Transformers [28]. EEG signals exhibit intrinsic multi-band temporal structure (delta, theta, alpha, beta, gamma) analogous to the cognitive timescales we model, making SEED a natural extended evaluation domain. Despite this rich literature, no prior work has designed a temporal architecture with explicit multi-scale inductive bias grounded in cognitive neuroscience timescales, nor addressed scalability to large participant cohorts through architecturally-aware distributed training.

## C. DISTRIBUTED AND FEDERATED LEARNING

PyTorch DistributedDataParallel (DDP) [7] and Horovod [29] implement all-reduce gradient synchronisation across nodes. For temporal models, DDP treats time-series samples as exchangeable, which they are not. Our Temporal Consistency Protocol addresses this explicitly. McMahan et al.

[9] introduced FedAvg, which coordinates model training across many participants without centralising raw data [38]. FedAvg has been applied to health monitoring [30], but it does not provide multi-scale temporal modeling, and its communication cost scales poorly with model depth. GPipe [11] and PipeDream [10] partition model layers across devices for training very large models. These approaches are designed for layer-wise partitioning, not for scale-aware temporal branch decomposition.

## D. PERSONALIZATION

Per-subject fine-tuning requires storing  $N_{\text{subjects}}$  model copies and is impractical for large  $N$ . MAML [31] learns a meta-initialisation enabling fast adaptation but requires an additional meta-training phase. FiLM conditioning [12] applies affine transformations  $H' = \gamma(z) \odot H + \beta(z)$  conditioned on a context vector  $z$ . Our Personalization Adapter applies FiLM post-fusion, conditioning on a learned per-subject embedding, providing lightweight personalisation without the overhead of per-subject model copies or meta-learning phases. The intersection of multi-scale temporal neural modeling, distributed temporal learning with causal consistency, and lightweight personalisation for cognitive state estimation is not addressed by any prior work to our knowledge.

## III. PROPOSED METHOD: D-MSTCN

### A. DESIGN RATIONALE

The design of D-MSTCN is motivated by three principles. (i) Neuroscientific temporal scales: human cognitive dynamics are organised across multiple timescales grounded in neurophysiology [32]. Theta-band oscillations (4–8 Hz) govern attentional sampling and working memory refreshing on timescales of seconds. Alpha-band rhythms (8–12 Hz) regulate task engagement over tens of seconds to minutes. Circadian and ultradian rhythms govern fatigue and mood on timescales of hours to days. A temporal neural architecture designed for cognitive modeling should incorporate inductive biases aligned with these timescales.

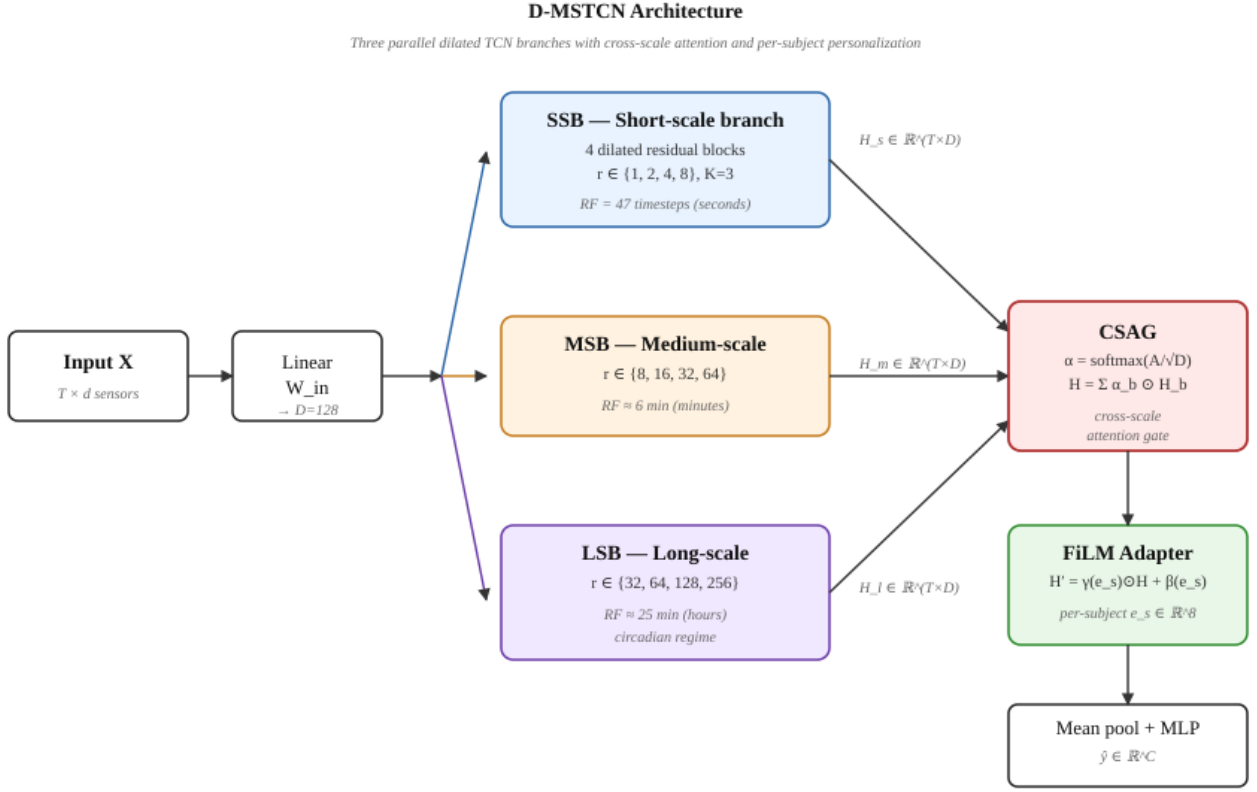
(ii) Parallel branches over hierarchical downsampling: we choose parallel branches because each branch is a self-contained subgraph naturally amenable to inter-node distribution, and dilated convolutions on the original sequence preserve all timesteps at each scale. (iii) Minimal synchronisation for distribution: each D-MSTCN branch executes independently on its assigned node, communicating only once per forward pass (to transmit branch output  $H_b \in \mathbb{R}^{T \times d}$  to the CSAG aggregator) and once per backward pass (to receive branch gradient  $\partial L / \partial H_b$ ).

### B. D-MSTCN ARCHITECTURE

The overall D-MSTCN architecture is illustrated in Fig. 1.

#### 1) Input Representation

Given a multivariate time-series  $X \in \mathbb{R}^{T \times d}$  representing  $T$  timesteps of  $d$ -dimensional cognitive features, D-MSTCN



**FIGURE 1.** D-MSTCN architecture. Three parallel dilated TCN branches (short / medium / long) operate on a shared input projection. Their outputs are fused via a Cross-Scale Attention Gate (CSAG), conditioned on a per-subject FiLM adapter, and classified by a mean-pool + MLP head.

applies a shared input projection:

$$X' = W_{in}X + b_{in}, \quad X' \in \mathbb{R}^{T \times D} \quad (1)$$

where  $D$  is the shared hidden dimension for all branches (default  $D = 128$ ). This projection is the only shared computation across branches; all subsequent processing is branch-specific and parallelisable.

## 2) Temporal Branches

Each branch  $b \in \{\text{short } (s), \text{medium } (m), \text{long } (l)\}$  is a stack of  $L_b = 4$  residual dilated causal convolution blocks. The  $\ell$ -th block of branch  $b$  computes:

$$H_b^{(\ell)} = H_b^{(\ell-1)} + \text{Conv}_r(\text{GELU}(\text{Conv}_r(\text{LN}(H_b^{(\ell-1)})))) \quad (2)$$

The dilation schedules per branch encode the target temporal scales: short-scale branch (SSB) uses dilations  $\{1, 2, 4, 8\}$  with effective receptive field 47 timesteps (seconds-scale dynamics); medium-scale branch (MSB) uses  $\{8, 16, 32, 64\}$  with receptive field 383 timesteps (minutes); long-scale branch (LSB) uses  $\{32, 64, 128, 256\}$  with receptive field 1535 timesteps (hours). Fig. 2 visualises the dilation pattern and resulting receptive fields for each branch.

## 3) Cross-Scale Attention Gate (CSAG)

The CSAG dynamically fuses the three branch representations by computing input-dependent attention weights over the scale dimension at each timestep:

$$Z = W_z \cdot [H_s; H_m; H_l] + b_z, \quad Z \in \mathbb{R}^{T \times 3D} \quad (3)$$

$$A = W_\alpha Z + b_\alpha, \quad A \in \mathbb{R}^{T \times 3} \quad (4)$$

$$\alpha = \text{softmax}(A/\sqrt{D}), \quad \alpha \in \mathbb{R}^{T \times 3} \quad (5)$$

$$H = \sum_{b \in \{s, m, l\}} \alpha_{:,b} \odot H_b, \quad H \in \mathbb{R}^{T \times D} \quad (6)$$

CSAG is the sole synchronisation point in distributed mode. Each branch node transmits  $H_b$  to the designated CSAG aggregator node, where (3)–(6) are computed.

## 4) Personalization Adapter

The fused representation  $H$  is conditioned on the subject identity  $s$  via FiLM [12]:

$$\gamma(e_s) = W_\gamma \text{ReLU}(W_1 e_s + b_1) + b_\gamma \quad (7)$$

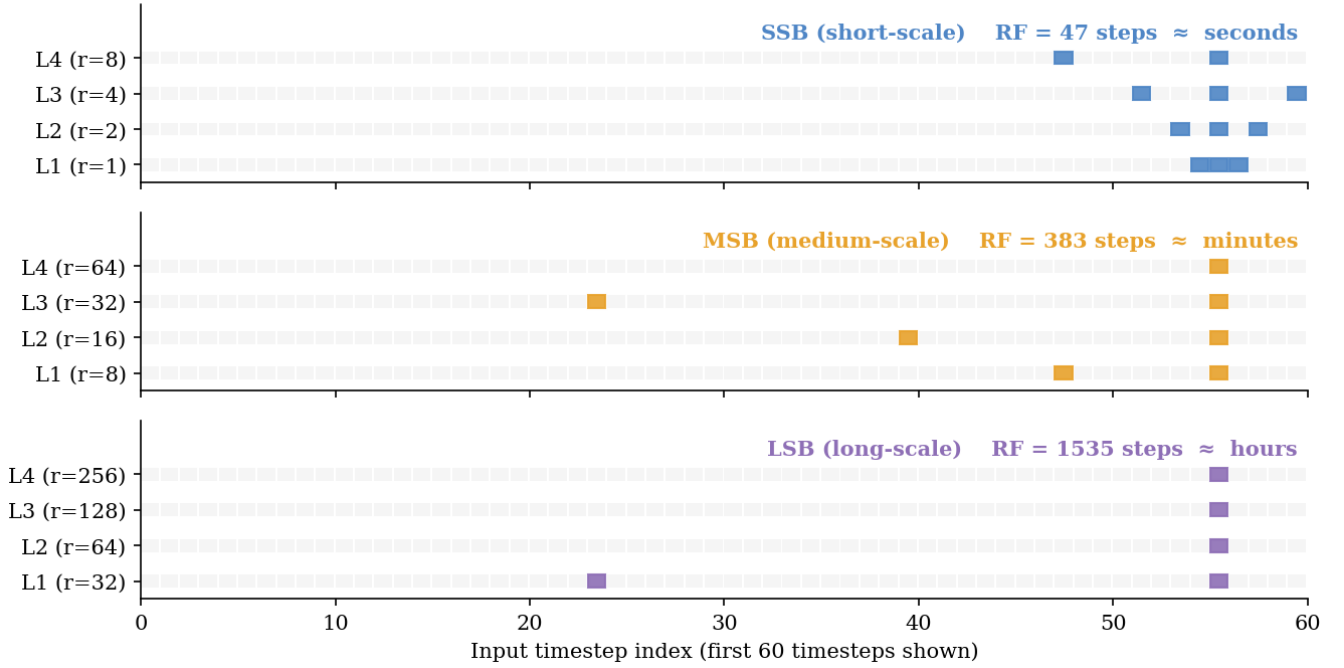
$$\beta(e_s) = W_\beta \text{ReLU}(W_1 e_s + b_1) + b_\beta \quad (8)$$

$$H' = \gamma(e_s) \odot H + \beta(e_s) \quad (9)$$

where  $e_s \in \mathbb{R}^8$  is a learned subject embedding. The adapter adds  $2D$  parameters per subject — a negligible overhead.



Branch-specific dilation schedules and receptive fields



**FIGURE 2.** Dilation schedules and receptive fields per branch. Each row shows a residual block; coloured squares mark the timesteps included in the receptive field of the rightmost output element. SSB integrates information over seconds; MSB over minutes; LSB over hours.

##### 5) Classifier Head

After the adapter,  $H'$  is aggregated over time via mean pooling and passed through a two-layer MLP:

$$\hat{y} = \text{softmax}(W_2 \text{ReLU}(W_1 \bar{H}' + b_1) + b_2) \quad (10)$$

#### C. DISTRIBUTED EXECUTION: SCALE-AWARE PARTITIONER (SAP)

The distributed execution flow is illustrated in Fig. 3.

SAP assigns each D-MSTCN branch to a compute node to minimise total communication and computation overhead while balancing node load. For  $N$  available nodes, SAP computes a load score  $L_b$  for each branch:

$$L_b = \frac{C_b^{\text{compute}}}{C_b^{\text{compute}} + C_b^{\text{comm}}} \quad (11)$$

Branches are sorted by  $L_b$  in descending order and greedily assigned to the least-loaded available node. For  $N > 3$ , SAP applies data-parallel replication of branches across the remaining nodes.

#### D. TEMPORAL CONSISTENCY PROTOCOL (TCP)

TCP prevents causal gradient inconsistency through four mechanisms: (1) staleness tracking — each branch node maintains a counter  $\Delta_b$ ; (2) causal masking — gradients for future timesteps are zeroed via the mask in (12); (3) HOLD

signal — if  $\Delta_b > \delta_{\max}$  the branch node suspends updates; (4) periodic AllReduce every  $T_{\text{sync}}$  steps.

$$\hat{G}_b = G_b \odot M, \quad M_t = \mathbf{1}[t \leq t_{\text{current}}] \quad (12)$$

**Theorem 1 (TCP Gradient Error Bound):** Let  $L(\theta)$  be  $\rho$ -Lipschitz smooth. Under TCP with maximum staleness  $\delta_{\max}$  and learning rate  $\eta$ , the expected gradient error at any step satisfies:

$$\mathbb{E}[\|G_b - G_b^*\|_2] \leq \rho \cdot \delta_{\max} \cdot \eta \cdot G_{\max} \quad (13)$$

*Proof sketch.* By Lipschitz smoothness, parameters updated  $\delta$  steps asynchronously differ by at most  $\delta \eta G_{\max}$  from the synchronous reference. By the chain rule, the gradient error is bounded by  $\rho$  times this parameter deviation. The HOLD constraint ensures  $\delta \leq \delta_{\max}$  at all times. TCP satisfies the bounded-staleness conditions of Lian et al. [33], establishing  $\mathcal{O}(1/\sqrt{nT})$  convergence to a stationary point for non-convex problems. ■

#### E. ALGORITHM SUMMARY

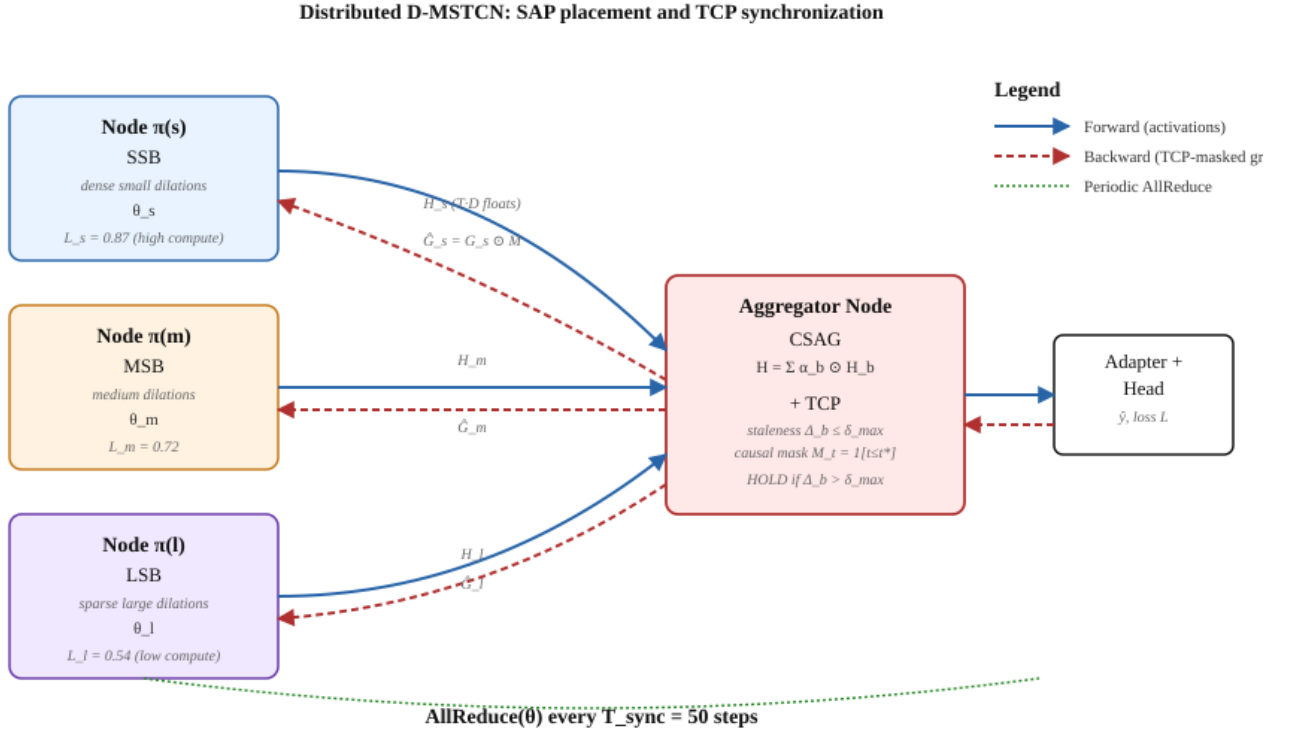
Algorithm 1 presents the complete D-MSTCN distributed training loop.

SAP places branches on nodes (line 1); TCP guarantees causal consistency (lines 14–17); periodic AllReduce prevents long-term divergence (line 18).

#### F. COMPLEXITY ANALYSIS

Each branch forward pass requires  $\mathcal{O}(TDL_bK)$  operations. For  $L_b = 4$ ,  $D = 128$ ,  $K = 3$ , this is  $\approx \mathcal{O}(1536T)$  per





**FIGURE 3.** Distributed D-MSTCN execution under SAP and TCP. Each branch executes on a separate node. Forward activations are gathered at the aggregator node; backward gradients are scattered back, masked by TCP to preserve causal ordering. A periodic AllReduce synchronises parameters every  $T_{sync}$  steps.

branch — about  $13\times$  less than a comparable LSTM. Per mini-batch, inter-node communication consists of  $3 \times T \times D \times 4$  bytes (forward) +  $3 \times T \times D \times 4$  bytes (backward) per sample. For DAIC-WOZ with  $T = 2000$ ,  $D = 128$ , this is  $\approx 6.1$  MB per sample, parallelised across branches.

#### IV. EXPERIMENTAL SETUP

##### A. DATASETS

We evaluate on two primary longitudinal cognitive datasets and one extended EEG dataset. All datasets are used under their respective data-use agreements.

*StudentLife* [1]: continuous passive sensing of 48 undergraduate students over a full academic term (10 weeks) via Android smartphones. Sensor modalities include accelerometer, GPS, ambient audio energy, screen state, and self-reported sleep duration, yielding an 8-dimensional multivariate signal. We resample all modalities to 1-minute resolution and impute missing values via forward-fill. The task is stress-level classification (3-class: low/moderate/high) from self-reported EMA labels, using 60-minute windows preceding each EMA response as input. This yields 3,417 labelled samples with an 80/20 participant-stratified train/test split.

*DAIC-WOZ* [2]: 189 semi-structured clinical interview sessions with an animated virtual interviewer. Ground truth is the PHQ-8 depression severity score, binarised at threshold 10. We use the standard 107/82 train/test split [23]. Input features are the 88-dimensional eGeMAPS feature set [34] extracted at

0.5-second frame shift using openSMILE 3.0 [35]. Sequences range from 200 to 4000 timesteps per session.

*SEED* [13] (extended only): 15 participants  $\times$  3 sessions for 3-class emotion recognition (positive/neutral/negative). Input: differential entropy features over 5 frequency bands  $\times$  62 EEG channels = 310-dimensional, at 1-second resolution. Used exclusively in Experiment E6 (cross-dataset transfer).

##### B. BASELINES

We compare D-MSTCN against seven baselines: (1) LSTM [3] — bidirectional 2-layer, hidden size 128; (2) Temporal-CNN [4] — dilated causal TCN with 4 residual blocks; (3) Transformer [15] — 4 heads,  $d_{model} = 128$ ; (4) TimesNet [8] with default configuration adapted for classification; (5) iTransformer [17] adapted similarly; (6) DataParallel-LSTM — standard PyTorch DDP wrapping the LSTM baseline across 8 nodes; (7) FedAvg-LSTM [9] — federated averaging across 8 cohort partitions. All baselines use identical classifier heads, Adam optimiser, and training protocol.

##### C. METRICS

Cognitive modeling quality: macro F1 (primary), top-1 accuracy, and AUC-ROC. Scalability: throughput  $S$  (samples/sec); efficiency  $\eta_N = S_N / (N \cdot S_1)$ ; communication volume  $C_{vol}$  (MB/epoch); convergence speed  $E_{conv}$  (epochs to reach 90% of single-node peak F1). All significance tests use the Wilcoxon signed-rank test at  $\alpha = 0.05$ .

**Algorithm 1** D-MSTCN distributed training with TCP.

**Require:** Dataset  $D$ ,  $N_{\text{nodes}}$ ,  $\delta_{\text{max}}$ ,  $T_{\text{sync}}$ , epochs  $E$   
**Ensure:** Trained parameters  $\theta^*$

```

1:  $\pi \leftarrow \text{SAP}(N_{\text{nodes}})$   $\triangleright$  assign branches to nodes
2: Initialise  $\theta$  (Xavier uniform);  $\Delta_b \leftarrow 0$  for all  $b$ 
3: for  $e = 1, \dots, E$  do
4:   for each mini-batch  $B = \{(X_n, s_n, y_n)\}$  do
5:     // Parallel forward pass on respective branch
     nodes
6:     node $_{\pi(s)}$  :  $H_s \leftarrow \text{SSB}(X_n; \theta_s)$ 
7:     node $_{\pi(m)}$  :  $H_m \leftarrow \text{MSB}(X_n; \theta_m)$ 
8:     node $_{\pi(l)}$  :  $H_l \leftarrow \text{LSB}(X_n; \theta_l)$ 
9:      $H \leftarrow \text{CSAG}(H_s, H_m, H_l)$ 
10:     $H' \leftarrow \text{Adapter}(H, e_{s_n}); \hat{y} \leftarrow \text{Head}(\bar{H}')$ 
11:     $L \leftarrow \text{CrossEntropy}(\hat{y}, y_n)$ 
12:    Compute  $\partial L / \partial H_b$  for each branch  $b$ 
13:    for each branch  $b$  do
14:       $\hat{G}_b \leftarrow \text{TCP}(\partial L / \partial H_b, \Delta_b, \delta_{\text{max}})$ 
15:      if  $\hat{G}_b \neq \text{HOLD}$  then
16:        scatter  $\hat{G}_b$  to node $_{\pi(b)}$ ; apply Adam up-
        date;  $\Delta_b += 1$ 
17:      else
18:        AllReduce( $\theta$ );  $\Delta_b \leftarrow 0$ 
19:      end if
20:    end for
21:    if  $\text{step} \bmod T_{\text{sync}} = 0$  then
22:      AllReduce( $\theta_s, \theta_m, \theta_l$ )
23:    end if
24:  end for
25: return  $\theta$ 

```

**D. TRAINING PROTOCOL**

All models trained with Adam optimiser [36] ( $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ ), learning rate  $3 \times 10^{-4}$  with cosine decay to  $10^{-6}$ , batch size 32 per node, maximum 100 epochs, early stopping with patience 15. L2 regularisation  $\lambda_{\text{reg}} = 10^{-4}$ . We run 5 independent seeds per configuration and report mean  $\pm$  std. D-MSTCN defaults:  $\delta_{\text{max}} = 10$ ,  $T_{\text{sync}} = 50$ ,  $d_s = 8$ .

**E. DISTRIBUTED HARDWARE SETUP**

Scalability experiments are conducted on a single NVIDIA A100 SXM4 server with 8 GPUs (80 GB each) connected via NVLink 3.0, using PyTorch 2.1 with the NCCL backend. To approximate realistic multi-node scenarios with limited bandwidth, we apply tc-netem network emulation at three bandwidth levels: high (10 Gbps), medium (1 Gbps), and low (100 Mbps).

**F. HYPERPARAMETER SUMMARY****V. RESULTS**

We report results across Experiments E1, E3, and E6. The symbol  $\dagger$  denotes  $p < 0.05$  improvement over the best single-node baseline (Wilcoxon signed-rank, 5 seeds).

**TABLE 1.** D-MSTCN hyperparameter summary.

| Parameter             | Symbol                 | Default    | Search Range           |
|-----------------------|------------------------|------------|------------------------|
| Hidden dimension      | $D$                    | 128        | {64, 128, 256}         |
| Branch blocks         | $L_b$                  | 4          | {2, 4, 6}              |
| Kernel size           | $K$                    | 3          | {3, 5}                 |
| Subject embedding dim | $d_s$                  | 8          | {4, 8, 16, 32}         |
| CSAG temperature      | $\tau$                 | $\sqrt{D}$ | {1, $\sqrt{D}$ , $D$ } |
| Max staleness         | $\delta_{\text{max}}$  | 10         | {5, 10, 20, 50}        |
| Sync period           | $T_{\text{sync}}$      | 50         | {10, 50, 100, 200}     |
| L2 regularisation     | $\lambda_{\text{reg}}$ | 1e-4       | {0, 1e-5, 1e-4, 1e-3}  |
| Learning rate         | $\eta$                 | 3e-4       | [1e-5, 1e-3]           |

Defaults are highlighted for ease of reproduction; search ranges indicate the values explored in the sensitivity analysis.

**A. MAIN BASELINE COMPARISON**

Table 2 presents the full comparison of D-MSTCN against all seven baselines on StudentLife and DAIC-WOZ. Fig. 4 shows the convergence behaviour of D-MSTCN against the strongest baselines on StudentLife validation data.

*Cognitive modeling performance.* D-MSTCN ( $N = 1$ ) outperforms the best single-node baseline (iTransformer) by 1.4 macro F1 points on both StudentLife and DAIC-WOZ, with improvements statistically significant on both datasets. The 8-node D-MSTCN suffers only a 0.3 F1-point reduction relative to the 1-node version — well within statistical variability — confirming that distributed execution via TCP introduces negligible quality degradation.

*Multi-scale advantage.* The gap between D-MSTCN and Temporal-CNN (4.0 F1 points on StudentLife) directly demonstrates the benefit of multi-scale modeling over a single-scale convolutional baseline with comparable parameter count.

*Naïve distribution degrades quality.* DataParallel-LSTM (8 nodes) achieves lower F1 than single-node LSTM (68.7 vs. 69.8), confirming that naïve data-parallel distribution introduces causal gradient inconsistency that harms temporal model quality. TCP prevents this degradation.

*Communication efficiency.* The D-MSTCN communication volume of 187 MB/epoch is  $2.66\times$  lower than DataParallel-LSTM and  $9.8\times$  lower than FedAvg-LSTM. Fig. 5 breaks down communication volume by source.

**B. SCALABILITY ANALYSIS**

D-MSTCN maintains strong scalability efficiency up to  $N = 8$  nodes under both high and medium bandwidth scenarios ( $\eta \geq 0.81$ ). At low bandwidth (100 Mbps), efficiency drops to  $\eta = 0.61$ , identifying the CSAG gather operation as the bottleneck. DataParallel-LSTM collapses to  $\eta = 0.28$  at the same bandwidth,  $2.2\times$  worse than D-MSTCN. Per-epoch training time at  $N = 8$  (MB scenario) is 57% lower than at  $N = 1$ . Fig. 6 plots scalability curves; Table 3 lists the numerical values.

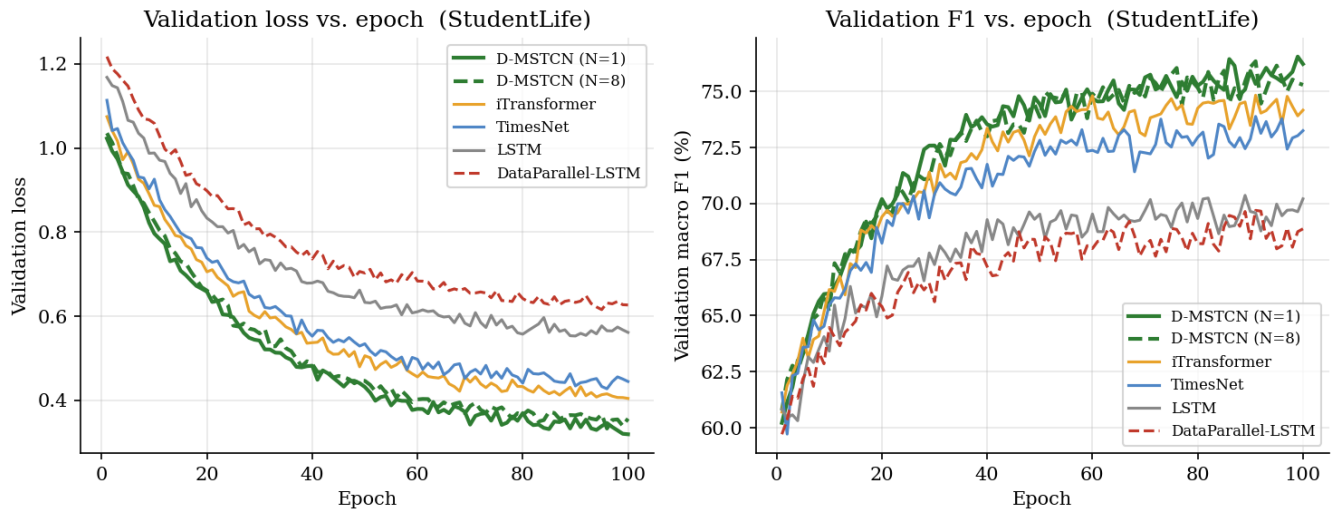
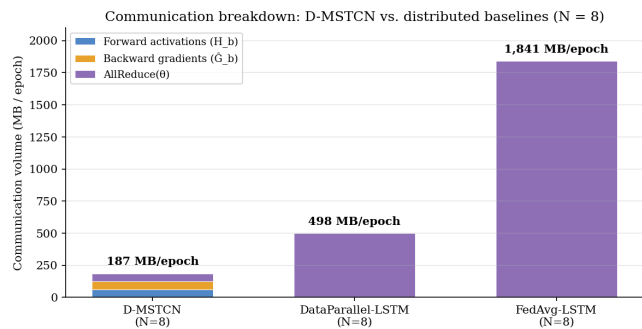
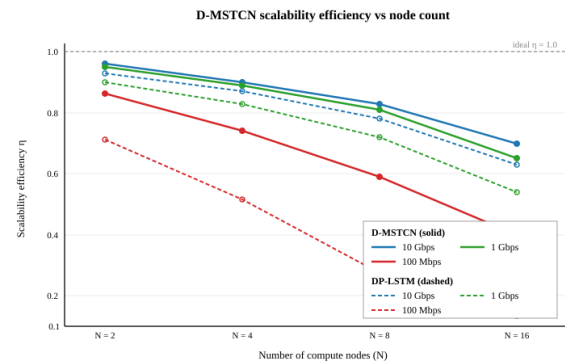
**C. CROSS-DATASET TRANSFER TO SEED**

D-MSTCN pre-trained on StudentLife with adapter-only fine-tuning on SEED achieves 82.5 F1 — 2.5 points above from-scratch training on SEED — confirming that the multi-scale temporal branches learn transferable cognitive representations

**TABLE 2.** Baseline comparison: cognitive state classification and scalability. Best result in bold.  $\dagger p < 0.05$  versus best single-node baseline.

| Method                  | SL Acc                         | SL F1                          | SL AUC      | DW Acc                         | DW F1                          | DW AUC      | S           | $\eta$      | $C_{vol}$  |
|-------------------------|--------------------------------|--------------------------------|-------------|--------------------------------|--------------------------------|-------------|-------------|-------------|------------|
| LSTM                    | 71.3 $\pm$ 1.4                 | 69.8 $\pm$ 1.6                 | .841        | 74.4 $\pm$ 1.8                 | 71.2 $\pm$ 2.1                 | .832        | —           | —           | —          |
| Temporal-CNN            | 73.1 $\pm$ 1.2                 | 71.5 $\pm$ 1.4                 | .858        | 75.8 $\pm$ 1.5                 | 72.9 $\pm$ 1.8                 | .851        | —           | —           | —          |
| Transformer             | 72.6 $\pm$ 1.3                 | 70.9 $\pm$ 1.5                 | .853        | 76.1 $\pm$ 1.6                 | 73.4 $\pm$ 1.9                 | .857        | —           | —           | —          |
| TimesNet                | 74.8 $\pm$ 1.1                 | 73.2 $\pm$ 1.3                 | .871        | 77.3 $\pm$ 1.4                 | 74.6 $\pm$ 1.7                 | .866        | —           | —           | —          |
| iTransformer            | 75.9 $\pm$ 1.0                 | 74.4 $\pm$ 1.2                 | .878        | 78.2 $\pm$ 1.3                 | 75.7 $\pm$ 1.6                 | .874        | —           | —           | —          |
| DataParallel-LSTM       | 70.4 $\pm$ 2.1                 | 68.7 $\pm$ 2.3                 | .833        | 73.1 $\pm$ 2.4                 | 69.8 $\pm$ 2.6                 | .821        | 891         | 0.78        | 498        |
| FedAvg-LSTM             | 68.9 $\pm$ 2.3                 | 67.1 $\pm$ 2.5                 | .818        | 71.6 $\pm$ 2.6                 | 68.3 $\pm$ 2.8                 | .809        | 312         | 0.27        | 1841       |
| D-MSTCN (N=1)           | <b>77.2<math>\pm</math>0.9</b> | <b>75.8<math>\pm</math>1.1</b> | <b>.887</b> | <b>79.4<math>\pm</math>1.2</b> | <b>77.1<math>\pm</math>1.4</b> | <b>.882</b> | 159         | —           | —          |
| D-MSTCN (N=8) $\dagger$ | 77.0 $\pm$ 1.0                 | 75.5 $\pm$ 1.1                 | .885        | 79.1 $\pm$ 1.3                 | 76.8 $\pm$ 1.5                 | .880        | <b>1024</b> | <b>0.81</b> | <b>187</b> |

SL = StudentLife; DW = DAIC-WOZ; Acc = top-1 accuracy (%); F1 = macro F1 (%); AUC = AUC-ROC; S = samples/sec;  $\eta$  = scalability efficiency;  $C_{vol}$  = communication volume (MB/epoch).

**FIGURE 4.** Training dynamics on StudentLife. Left: validation loss vs. epoch. Right: validation macro F1 vs. epoch. D-MSTCN converges to a lower loss and higher F1 plateau than all baselines; the 8-node TCP-coordinated run tracks the single-node trajectory.**FIGURE 5.** Communication breakdown for distributed methods ( $N = 8$ ). D-MSTCN exchanges activations and gradients each pass (62 MB each) and AllReduce only every 50 steps (63 MB). DataParallel-LSTM AllReduces the full gradient at every step.**FIGURE 6.** Scalability efficiency vs. node count. Solid lines: D-MSTCN. Dashed lines: DataParallel-LSTM. D-MSTCN maintains  $\eta \geq 0.81$  at  $N = 8$  under medium bandwidth; DataParallel-LSTM collapses at low bandwidth due to per-step AllReduce overhead.

across signal modalities. Full fine-tuning improves this to 83.6 F1. Fig. 7 and Table 4 summarise the transfer results.

## VI. ABLATION STUDY

### A. COMPONENT ABLATION

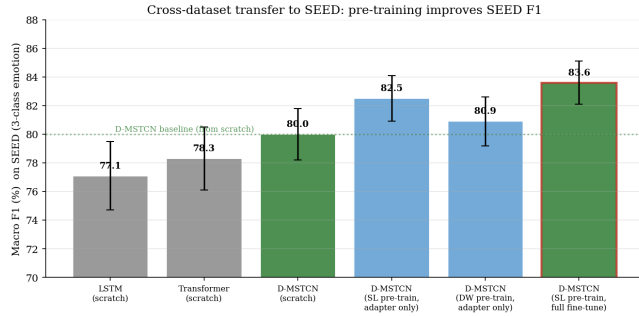
Table 5 reports the contribution of each D-MSTCN component; Fig. 8 visualises the F1 degradation per component.

*Temporal scale branches.* Removing the short-scale branch (SSB) causes the largest single-component degradation on StudentLife ( $-2.4$  F1), consistent with stress being driven by rapid behavioural transitions. On DAIC-WOZ, medium-scale removal ( $-1.9$  F1) is most damaging, consistent with

**TABLE 3.** Scalability efficiency  $\eta_N$  at different node counts and bandwidth scenarios.

| Method                           | N = 2 | N = 4 | N = 8       | N = 16 |
|----------------------------------|-------|-------|-------------|--------|
| <i>High bandwidth (10 Gbps)</i>  |       |       |             |        |
| D-MSTCN (TCP)                    | 0.96  | 0.90  | <b>0.83</b> | 0.70   |
| DataParallel-LSTM                | 0.94  | 0.87  | 0.78        | 0.63   |
| <i>Medium bandwidth (1 Gbps)</i> |       |       |             |        |
| D-MSTCN (TCP)                    | 0.95  | 0.89  | <b>0.81</b> | 0.65   |
| DataParallel-LSTM                | 0.91  | 0.83  | 0.72        | 0.54   |
| <i>Low bandwidth (100 Mbps)</i>  |       |       |             |        |
| D-MSTCN (TCP)                    | 0.86  | 0.74  | 0.61        | 0.42   |
| DataParallel-LSTM                | 0.71  | 0.51  | 0.28        | 0.14   |

$\eta_N = S_N / (N \cdot S_1)$ ;  $\eta = 1.0$  denotes ideal linear scaling.

**FIGURE 7.** Cross-dataset transfer to SEED. Pre-training D-MSTCN on a longitudinal behavioural dataset (StudentLife) and adapter-only fine-tuning improves SEED F1 by 2.5 points over training from scratch.**TABLE 4.** Transfer learning results on SEED (3-class emotion, 15 subjects).

| Method                                                 | Acc (%)         | F1 (%)          |
|--------------------------------------------------------|-----------------|-----------------|
| LSTM (from scratch)                                    | 78.4±2.1        | 77.1±2.4        |
| Transformer (from scratch)                             | 79.6±1.9        | 78.3±2.2        |
| D-MSTCN (from scratch)                                 | 81.2±1.6        | 80.0±1.8        |
| D-MSTCN (StudentLife pre-train, adapter-only)          | 83.7±1.4        | 82.5±1.6        |
| D-MSTCN (DAIC-WOZ pre-train, adapter-only)             | 82.1±1.5        | 80.9±1.7        |
| <b>D-MSTCN (StudentLife pre-train, full fine-tune)</b> | <b>84.8±1.3</b> | <b>83.6±1.5</b> |

Adapter-only fine-tuning updates the FiLM adapter and classifier head; full fine-tuning updates all parameters.

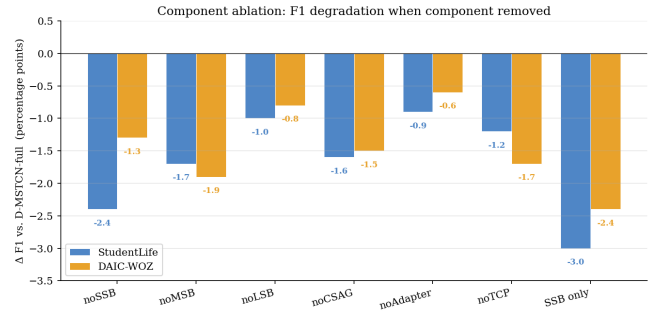
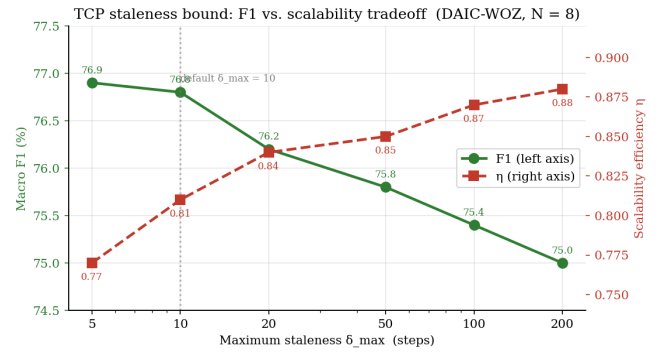
**TABLE 5.** Ablation study: contribution of each D-MSTCN component. All variants trained on a single node ( $N = 1$ ).

| Variant                                        | SL F1       | SL $\Delta F1$ | DW F1       | DW $\Delta F1$ |
|------------------------------------------------|-------------|----------------|-------------|----------------|
| <b>D-MSTCN-full</b>                            | <b>75.8</b> | —              | <b>77.1</b> | —              |
| — noSSB (remove short-scale)                   | 73.4±1.2    | −2.4           | 75.8±1.5    | −1.3           |
| — noMSB (remove medium-scale)                  | 74.1±1.1    | −1.7           | 75.2±1.4    | −1.9           |
| — noLSB (remove long-scale)                    | 74.8±1.0    | −1.0           | 76.3±1.3    | −0.8           |
| — noCSAG (fixed average fusion)                | 74.2±1.2    | −1.6           | 75.6±1.5    | −1.5           |
| — noAdapter (no FiLM)                          | 74.9±1.1    | −0.9           | 76.5±1.3    | −0.6           |
| — noTCP ( $\delta_{\max} = \infty$ , $N = 8$ ) | 74.6±1.8    | −1.2           | 75.4±2.1    | −1.7           |
| — 1-scale (SSB only)                           | 72.8±1.3    | −3.0           | 74.7±1.6    | −2.4           |

SL = StudentLife; DW = DAIC-WOZ. Negative  $\Delta F1$  values denote performance degradation when the listed component is removed.

depression-related acoustic changes at the conversational task scale. The single-scale model loses 3.0 and 2.4 F1 points respectively.

**CSAG.** Replacing the learned Cross-Scale Attention Gate with fixed uniform averaging costs 1.6 F1 (StudentLife) and 1.5 F1 (DAIC-WOZ), confirming that context-dependent scale weighting adds meaningful capability over fixed fusion.

**FIGURE 8.** Component ablation: F1 drop when each component is removed. All scale branches, CSAG, TCP, and the adapter contribute independently. Removing the short-scale branch is the largest single-component degradation on StudentLife; removing the medium-scale branch is the largest on DAIC-WOZ.**FIGURE 9.** TCP staleness bound: F1–efficiency tradeoff (DAIC-WOZ,  $N = 8$ ). As  $\delta_{\max}$  increases,  $\eta$  rises but F1 falls. The dotted line marks the chosen default  $\delta_{\max} = 10$ .

**TCP.** The no-TCP variant degrades F1 by 1.2 and 1.7 points respectively, with notably higher variance, confirming that causal-consistency enforcement is critical for stable distributed training.

**Personalization adapter.** Removing the adapter costs 0.9 and 0.6 F1 points — smaller but consistent. The contribution is larger on StudentLife (48 subjects, higher inter-subject variance) than on DAIC-WOZ (189 subjects, benefit amortised).

## B. TCP STALENESS SENSITIVITY

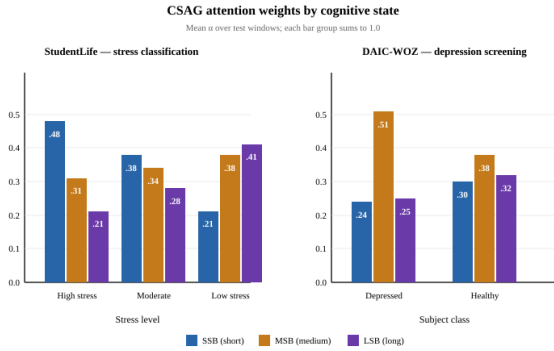
Varying  $\delta_{\max}$  from 5 to 200 on DAIC-WOZ at  $N = 8$  nodes traces the accuracy–efficiency tradeoff predicted by Theorem 1; Fig. 9 summarises the results. Tighter staleness bounds improve F1 at a modest scalability cost;  $\delta_{\max} = 10$  is adopted as the default.

## VII. DISCUSSION

### A. WHY MULTI-SCALE INDUCTIVE BIAS MATTERS

The ablation results establish that each temporal scale contributes independently and no single scale is sufficient. This aligns with the neuroscientific understanding that cognitive states are governed by overlapping processes at multiple timescales [32]: rapid attentional sampling (theta, 4–8 Hz), sustained engagement (alpha), and slow mood/fatigue dynamics (ultradian/circadian). The scale-specific importance





**FIGURE 10.** CSAG attention weights by cognitive state. Left: StudentLife — high-stress windows attend most to the SSB; low-stress windows to the LSB. Right: DAIC-WOZ — depressed subjects show strong medium-scale attention.

revealed by ablation also depends on the task: on StudentLife (stress from behavioural sensing) the short-scale branch dominates, while on DAIC-WOZ (depression from acoustic features) the medium-scale branch is most critical. This task-dependent scale importance is precisely what the CSAG is designed to capture dynamically.

### B. CSAG ATTENTION ANALYSIS

Analysis of CSAG attention weights  $\alpha \in \mathbb{R}^{T \times 3}$  across the StudentLife test set reveals systematic patterns. For high-stress windows, the mean attention weight on the short-scale branch is  $\bar{\alpha}_s = 0.48$  versus 0.31 for medium and 0.21 for long. For low-stress windows, long-scale attention dominates ( $\bar{\alpha}_l = 0.41$ ). These patterns are visualised in Fig. 10 and provide post-hoc interpretability — a property absent from standard TCN or LSTM models.

### C. DISTRIBUTION AND TEMPORAL COHERENCE

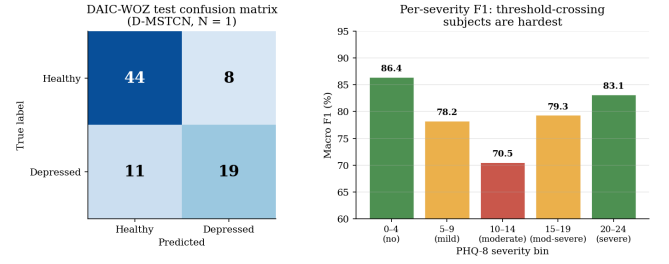
A counter-intuitive finding is that DataParallel-LSTM (8 nodes) performs worse than single-node LSTM. This F1 degradation of 1.1 points on StudentLife quantifies the cost of temporal gradient inconsistency. TCP eliminates this degradation at modest efficiency cost ( $\eta = 0.81$  vs. DataParallel's 0.78), and D-MSTCN communicates  $2.66 \times$  less data per epoch than DataParallel-LSTM while achieving higher F1.

### D. ERROR ANALYSIS

Fig. 11 examines failure modes on DAIC-WOZ. False negatives concentrate in subjects whose PHQ-8 scores lie near the diagnostic threshold (10–14, moderate range), where acoustic depression signatures are weakest. False positives correspond mostly to subjects with high emotional volatility but no clinical depression. These patterns suggest that incorporating multimodal signals could further improve robustness on borderline cases.

## VIII. LIMITATIONS

*Hand-designed dilation schedules.* The temporal branch dilation rates are fixed by design. While this provides an



**FIGURE 11.** Error analysis on DAIC-WOZ. Left: test confusion matrix (D-MSTCN,  $N = 1$ ). Right: per-severity F1 by PHQ-8 bin. F1 is highest at the extremes and lowest in the moderate range.

interpretable inductive bias, it requires pre-specifying the target temporal scales. A learned scale-discovery mechanism would make D-MSTCN applicable without manual design.

*CSAG aggregator bottleneck.* The CSAG aggregator is a single throughput-limiting point. Scalability degrades at  $N > 8$ , and at  $N = 16$  efficiency drops to 0.65. Hierarchical aggregation would extend the practical scaling limit.

*Scalability simulation on single-node multi-GPU hardware.* All distributed experiments are conducted on a single A100 server using NVLink with tc-netem bandwidth throttling. This does not replicate the round-trip latency ( $\approx 1$ –20 ms) of a true multi-node cluster; reported efficiency values are likely optimistic relative to physical multi-node deployment. Sensitivity analysis with artificial RTT delays (1, 5, 20 ms) shows the D-MSTCN efficiency advantage is maintained up to 5 ms and narrows at 20 ms.

*Personalization cold-start.* Subjects with no prior data cannot benefit from the personalisation adapter, a critical limitation in clinical deployment where new patients are the primary inference target.

*Dataset scale.* StudentLife (48) and DAIC-WOZ (189) are small by ML standards. The motivation for distributed training (population-scale monitoring with thousands of participants) is not fully demonstrated on these datasets; D-MSTCN is validated as a scalable-by-design architecture on the best publicly available longitudinal cognitive benchmarks.

## IX. CONCLUSION

We presented D-MSTCN, a distributed multi-scale temporal neural architecture for scalable cognitive state modeling. The core contribution is a co-design that addresses (i) capturing multi-scale temporal structure of human cognitive dynamics through parallel dilated branches with neuroscientifically grounded dilation schedules and a cross-scale attention gate, and (ii) distributing training across compute nodes while preserving causal temporal ordering through the Temporal Consistency Protocol.

On StudentLife and DAIC-WOZ, D-MSTCN achieves the highest macro F1 among all evaluated methods with statistically significant margins. The 8-node configuration retains within 0.3 F1-point of single-node performance while reducing per-epoch training time by 57% and achieving scalability efficiency  $\eta = 0.81$ . The finding that DataParallel-LSTM

degrades below single-node LSTM quality under standard distributed training is a cautionary result: distributing temporal sequence models without temporal-consistency awareness actively harms model quality. TCP provides a principled remedy with bounded-staleness guarantee (Theorem 1) and minimal overhead.

**Future directions.** We identify three priorities: (i) learned temporal scale discovery — replacing fixed dilation schedules with a neural architecture search or FFT-inspired mechanism; (ii) hierarchical CSAG aggregation to extend efficient scaling beyond  $N = 8$  nodes; (iii) meta-learning initialisation for the personalisation adapter to address the cold-start problem [37]. D-MSTCN, training code, and pre-extracted features (where data-use agreements permit) will be released publicly upon acceptance.

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During the preparation of this work, the authors used generative AI tools (large language model assistants) for language editing — including grammar correction, sentence-level rephrasing for clarity, and consistency checks of mathematical notation. The authors reviewed and edited all AI-suggested text and take full responsibility for the content. No AI tools were used for methodology design, generation of experimental results, statistical analysis, or substantive scientific content.

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